

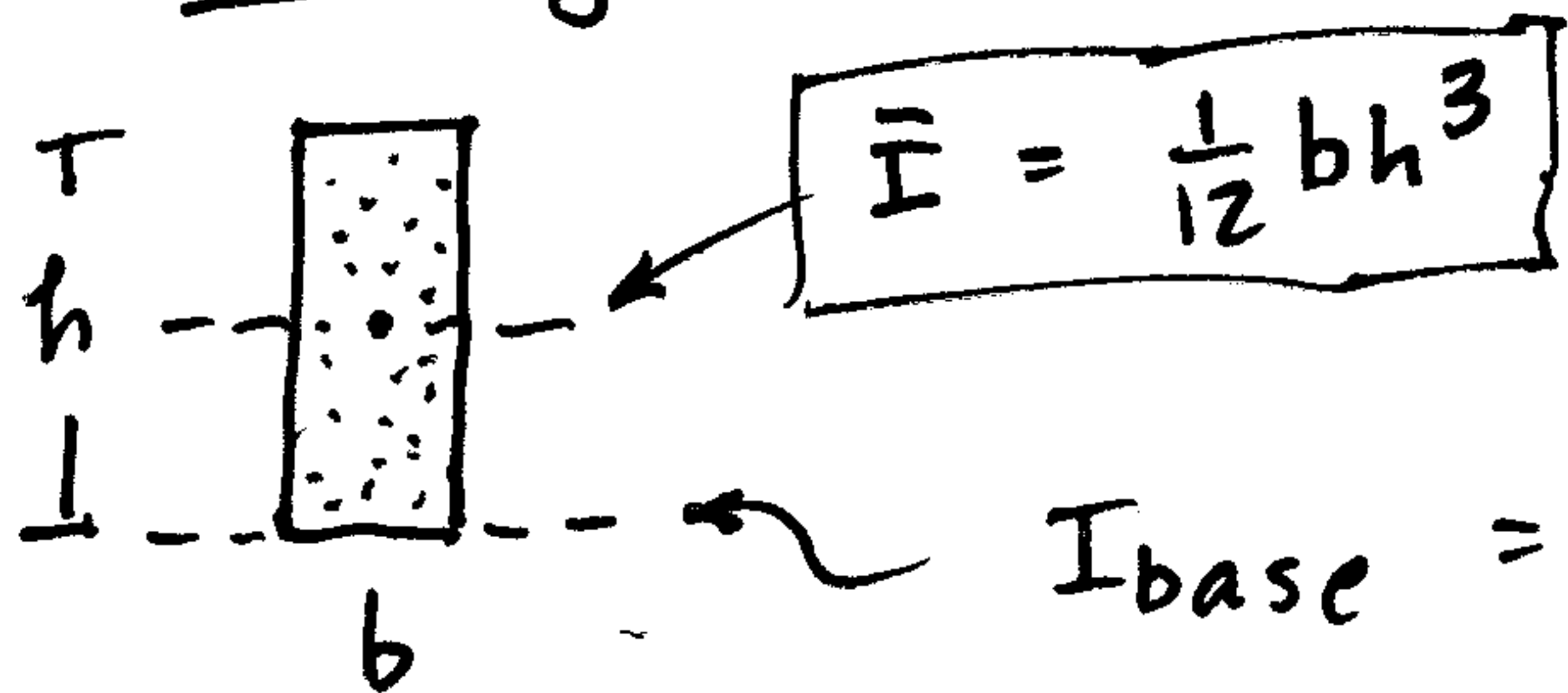
AREA MOI of Common Shapes

Computing I_{base} from $\bar{I}$ or vice versa.

① Rectangle

Parallel Axis Theorem

"PAT": $I_{Axis} = \bar{I} + Ad^2$



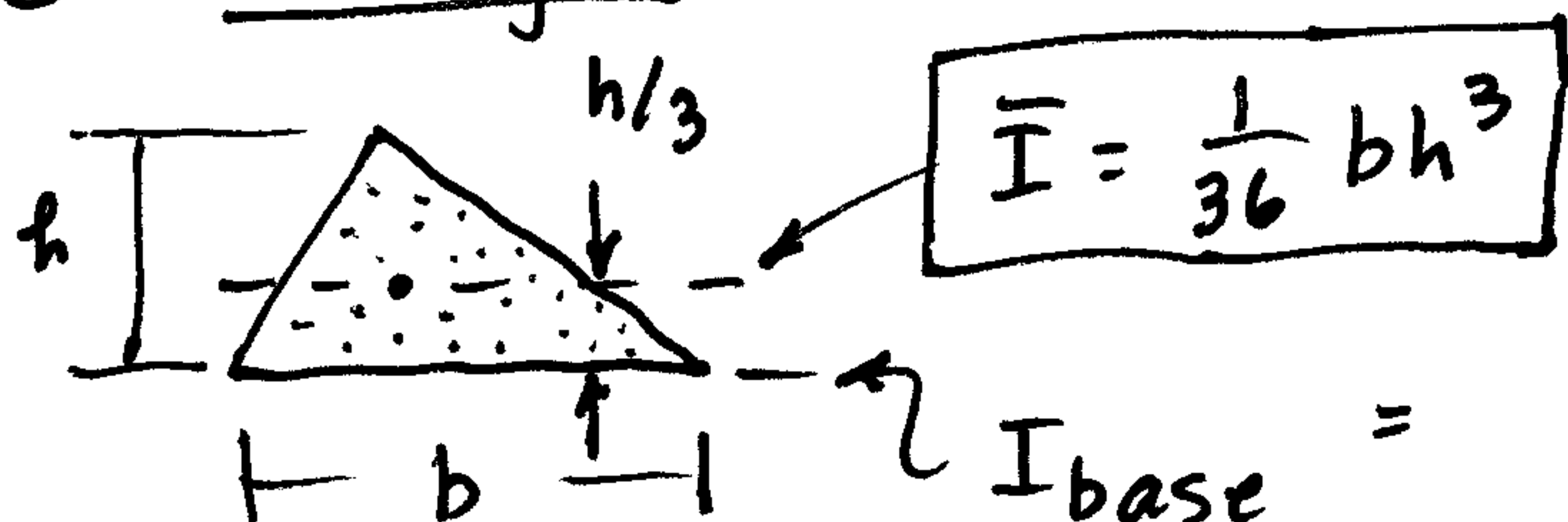
$$I_{base} = \left[\bar{I} + \underbrace{(b \cdot h)}_A \left(\underbrace{\frac{h}{2}}_d \right)^2 \right]$$

$$= bh^3 \left[\frac{1}{12} + \frac{1}{4} \right] = \frac{4}{12}bh^3 = \frac{1}{3}bh^3$$

$\frac{3}{12}$

$$I_{base} = \frac{1}{3}bh^3$$

② Triangle



$$I_{base} = \bar{I} + Ad^2$$

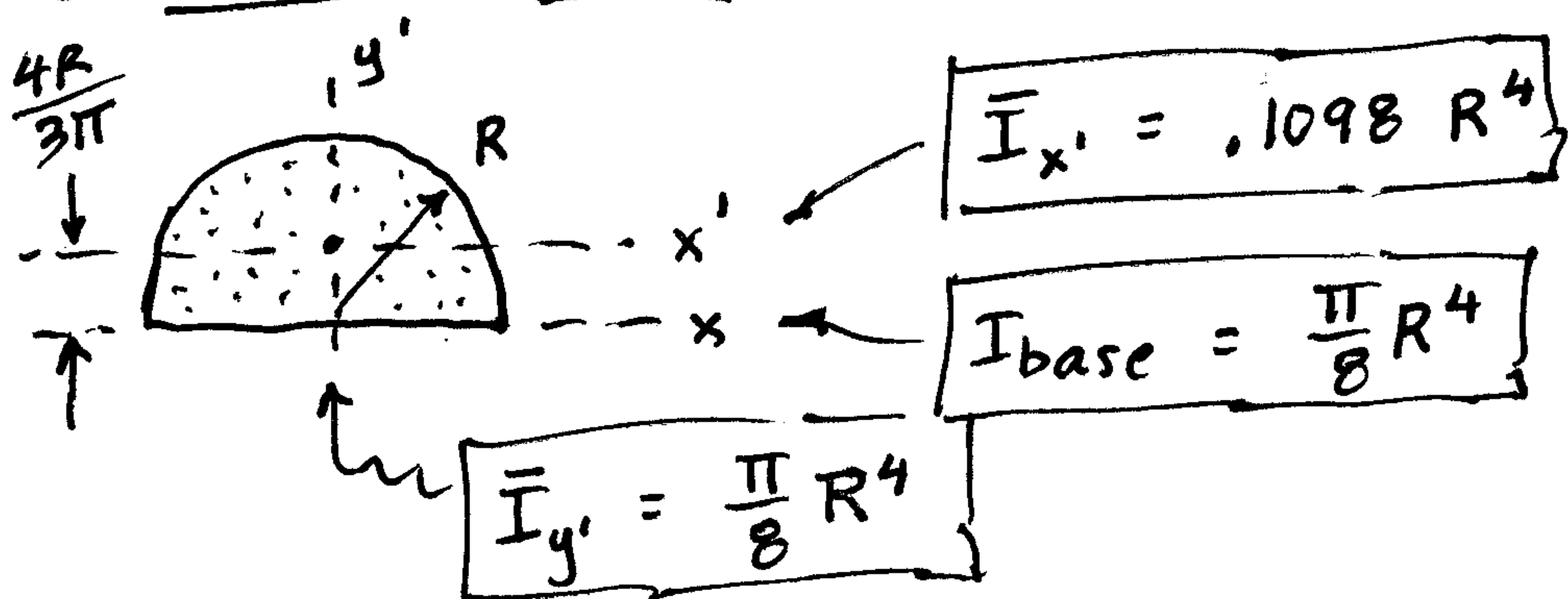
$$= \frac{1}{36}bh^3 + \left(\frac{1}{2}bh \right) \left(\frac{h}{3} \right)^2$$

$$= bh^3 \left[\frac{1}{36} + \frac{1}{18} \right] = \frac{3}{36}bh^3$$

$\frac{2}{36}$ $\frac{1}{12}bh^3$

$$I_{base} = \frac{1}{12}bh^3$$

③ Half Circle



$$\bar{I}_{x'} = .1098 R^4$$

$$I_{base} = \frac{\pi}{8} R^4$$

$$\bar{I}_{y'} = \frac{\pi}{8} R^4$$

Start w/ I_{base} , compute $\bar{I}_{x'}$:

$$I_{base} = \bar{I}_{x'} + Ad^2$$

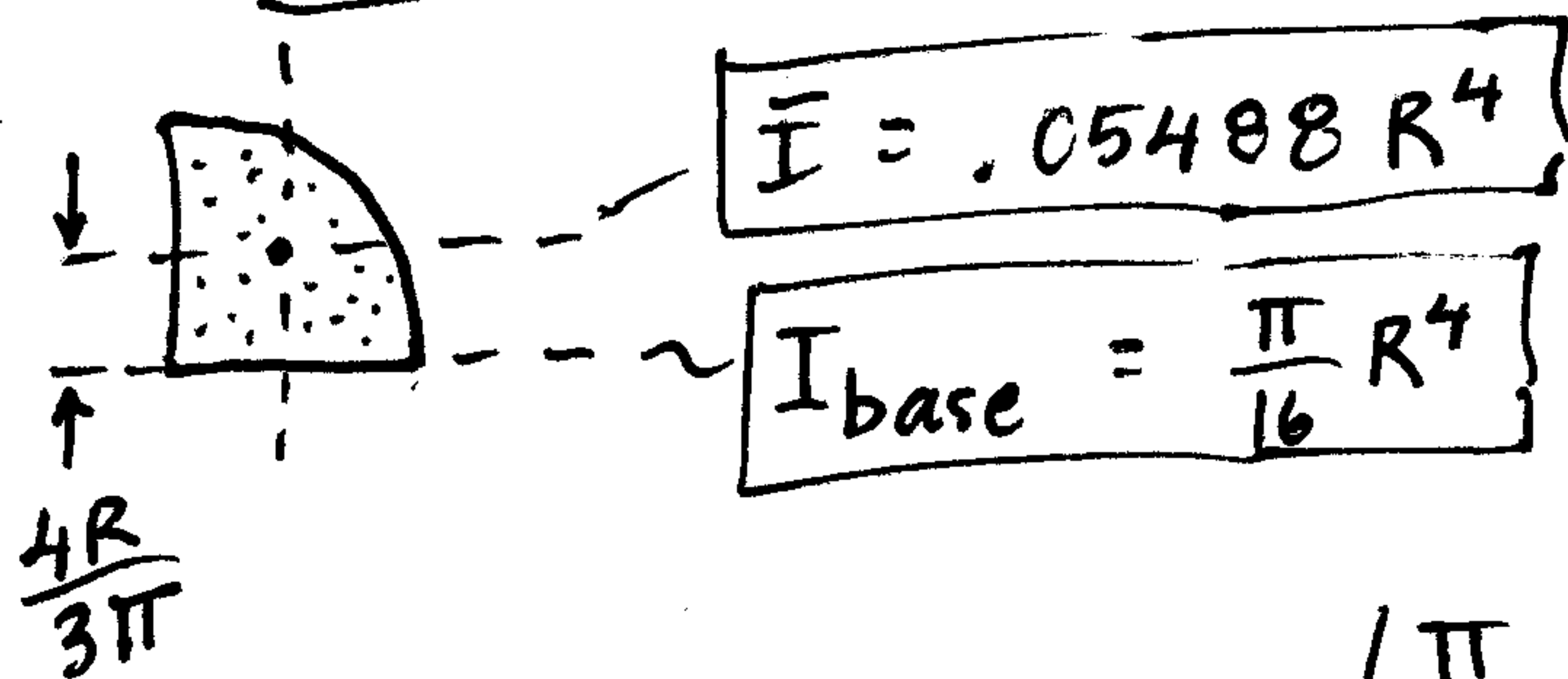
$$\bar{I}_{x'} = I_{base} - Ad^2$$

$$= \frac{\pi}{8} R^4 - \left(\frac{\pi R^2}{2} \right) \left(\frac{4R}{3\pi} \right)^2$$

$$= \left[\frac{\pi}{8} - \frac{16}{18\pi} \right] R^4$$

$$\bar{I}_{x'} = 0.10976 R^4 = .1098 R^4$$

④ Quarter Circle



$$\bar{I} = .05488 R^4$$

$$I_{base} = \frac{\pi}{16} R^4$$

$$I_{base} = \bar{I} + Ad^2$$

$$\bar{I} = I_{base} - Ad^2$$

$$= \frac{\pi}{16} R^4 - \left(\frac{\pi R^2}{4} \right) \left(\frac{4R}{3\pi} \right)^2$$

$$4 \cdot 9 = 36$$

$$= \left(\frac{\pi}{16} - \frac{16}{36\pi} \right) R^4$$

$$\bar{I} = .05488 R^4$$